Reverse Tabu: Pedagogical Foundation and Gameplay

1. Game Design and Mechanics

In the traditional Taboo game, the player tries to describe a target word without using forbidden words. Reverse Tabu works in the opposite way. The player is given only a set of keywords. Using these keywords, the player must explain the target concept directly without providing a formal definition. They are not allowed to form a definitional sentence. The only thing they need to do is to reveal the context of the target concept in their own mind using the given keywords. The main goal of the game is to establish accurate and meaningful connections between the keywords and the target concept, thereby expressing the concept in a holistic way.

For example, when the target concept is "router," the player is given keywords such as *network, traffic, IP address, routing, data packet*. The player explains what a router does without defining it directly, for instance: "It manages data traffic between different networks, looks at IP addresses, and directs packets to the correct destination." In this case, the player does not answer the question "What is a router?" directly but instead explains the function of the concept through the keywords.

2. Target Audience

This game is designed for high school students (ages 13+). The main reason for this choice is that the game involves abstract computer science concepts such as routers, IP addresses, data packets, and protocols. According to Piaget's theory of cognitive development, middle school students are mostly in the concrete operational stage and may have difficulty processing such abstract concepts. By the time students reach high school, they begin to transition into the formal operational stage. Therefore, these conceptual and relational thinking skills can develop more easily at this level.

3. Pedagogical Foundation

Constructivist Approach: According to Tezci and Köksalan (2007), in constructivism, learners are not passive recipients of knowledge but active constructors of it. In Reverse Tabu, the student does not memorize and repeat a ready-made definition. Instead, they transform the given keywords into a meaningful whole within their own mental schema. In this process, the student establishes connections between their existing knowledge and new information, thereby reconstructing their own understanding (Tezci & Köksalan, 2007, p. 73).

Meaningful Learning and Advance Organizers: According to Çakıcı and Altunay (2006), if a meaningful connection is not established between the new information to be learned and the prior knowledge existing in the mind, learning remains at the level of rote memorization. The keywords used in the game function similarly to Ausubel's concept of "advance organizers." These words create a bridge in the student's mind between the new concept and their existing knowledge. For instance, when a student hears the word "router," they recall previously encountered concepts like network, connection, and the internet. This allows them to meaningfully integrate the new information into their existing cognitive structure. Thus, the target concept ceases to be an isolated, easily forgotten piece of information and becomes part of a permanent, meaningful knowledge network (Çakıcı & Altunay, 2006, pp. 13-14).

References

Çakıcı, D. & Altunay, U. (2006). Advance Organizers and Employment in Teaching. *Kastamonu Education Journal*, 14(1), 13-14.

Tezci, E. & Köksalan, B. (2007). Constructivist Education and Constructivist Teaching Approaches. *Eastern Anatolia Region Research*, p. 73.